Junjie Nian 年骏杰

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ABOUT

Undergraduate researcher at Fudan University majoring in Artificial Intelligence (ranked 1st in class, GPA 3.93/4.0). Currently conducting research on large reasoning models, agents, and interpretability at FudanNLP ALEX Lab, advised by Prof. Yixin Cao. Published 4 papers, including 1 ACL 2026 Findings paper and 3 arXiv preprints. Will be an exchange student at UC San Diego in Fall 2026. Passionate about advancing the frontiers of AI reasoning and committed to a career in AI research.

EDUCATION

2024.9 – Present
Fudan University — B.Eng. in Artificial Intelligence
School of Computer Science (Honors Track). GPA 3.93/4.0, ranked 1st in major.

2026.9 – 2027.1
UC San Diego — Exchange Student
Computer Science & Engineering.

2021.9 – 2024.6
The High School Affiliated to Fudan University — Buqing Honors Class

RESEARCH EXPERIENCE

2025.10 – Present
FudanNLP ALEX Lab — Undergraduate Researcher
Advised by Prof. Yixin Cao. Research on Large Reasoning Model capabilities, agent systems, and mechanistic interpretability. Produced 4 papers (including 1 ACL 2026 Findings). Independently explored on CoT Folding and Multilingual Neuron Analysis projects. Currently leading a project on agent interpretability.

2025.3 – 2025.6
Embodied Intelligence Research Project — Tengfei Innovation Program
Advised by Prof. Shang Li. Built small-scale robots with asynchronous servo motor control and embedded development board integration.

PUBLICATIONS

Kang Chen[*], **Junjie Nian**[*], Yixin Cao, Yugang Jiang. "SliceGraph: Mapping Process Isomers in Multi-Run Chain-of-Thought Reasoning." **[arXiv]**

Kang Chen[*], Zhuoka Feng[*], Sihan Zhao, Kai Xiong, **Junjie Nian**, Yaoning Wang, Changyi Xiao, Yixin Cao. "NEX: Neuron Explore-Exploit Scoring for Label-Free Chain-of-Thought Selection and Model Ranking." **[arXiv]**

Kang Chen[*], Fan Yu[*], **Junjie Nian**, Shihan Zhao, Zhuoka Feng, Zijun Yao, Heng Wang, Minshen Yu, Yixin Cao. "Thinking Traps in Long Chain-of-Thought: A Measurable Study and Trap-Aware Adaptive Restart." **Findings of ACL, 2026.** **[arXiv]**

Zhuoka Feng[*], Kang Chen[*], Sihan Zhao[†], Kai Xiong[†], Yaoning Wang[†], Minshen Yu, **Junjie Nian**, Changyi Xiao, Yixin Cao, Yugang Jiang. "ARM: Role-Conditioned Neuron Transplantation for Training-Free Generalist LLM Agent Merging." **[arXiv]**

* Equal contribution † Equal second-author contribution

HONORS & AWARDS

Duckbill Special Scholarship, Fudan University (equivalent to First-Class Outstanding Student Scholarship)	**2025**
Third Prize, China Undergraduate Mathematical Contest in Modelling, Shanghai	**2025**
Third Prize, The Chinese Mathematics Competitions for College Students, Shanghai	**2025**
Freshman Scholarship, Fudan University (New Engineering Discipline)	**2025**
First Prize, Tengfei College Entrance Exam Freshman Scholarship, Fudan University	**2024**

SELECTED PROJECTS

CoT Folding: Structural Analysis of LLM Reasoning Trajectories
Framework treating Chain-of-Thought as analogous to protein folding. Measures pairwise similarity between token segments using neuron activation patterns. Proposes Native Fold Score (NFS), a fully unsupervised, parameter-free quality metric for CoT reasoning.

Multilingual Neuron Analysis for LLM Reasoning
Extended NAD framework to multilingual settings across 18 languages and 2,250 math reasoning samples. Found that quality reasoning neurons are 99.8% language-agnostic; moderate code-switching outperforms monolingual reasoning by +68.8%.

80/20 Rule of Token Entropy in LLM Reasoning
Independent reproduction and extension of Qwen team's NeurIPS 2025 work with five-metric comparison, multilingual CoT analysis, and entropy-frequency correlation study across 539K+ samples and 70M+ tokens.

Paper Assistant System
Full-stack academic platform: AI-powered LaTeX editor (GLM-4 feedback, real-time compilation, Monaco with SyncTeX) and ArXiv Paper Agent with automated daily scoring. React + FastAPI.

SKILLS

Programming: Python, C/C++

AI / Research: PyTorch, NumPy/SciPy, neuron-level interpretability analysis, data analysis & visualization

Engineering: React, FastAPI, Three.js, Docker, LaTeX

Math & Physics: National Physics Olympiad Second Prize (x2); full marks in all core math & physics courses

ENGLISH PROFICIENCY

TOEFL **111** CET-6 **619** CET-4 **636**